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Course: **Java FSE Digital Nurture**

Week 1

Exercise 1: Singleton Pattern

Aim

To implement the Singleton Design Pattern in Java.

Theory

The Singleton Pattern is a Creational Design Pattern that ensures only one instance of a class exists and provides a global point of access to it.

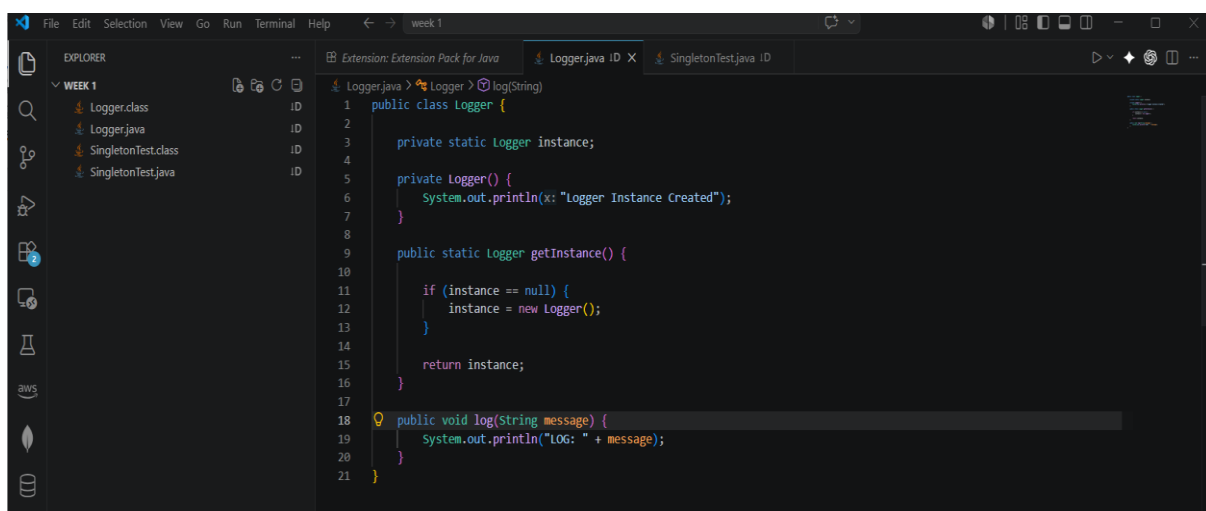
Algorithm

1. Create a class Logger.
2. Make its constructor private.
3. Create a private static instance variable.
4. Create a public static method getInstance().
5. Return the same instance whenever called.
6. Test using multiple references.

Program

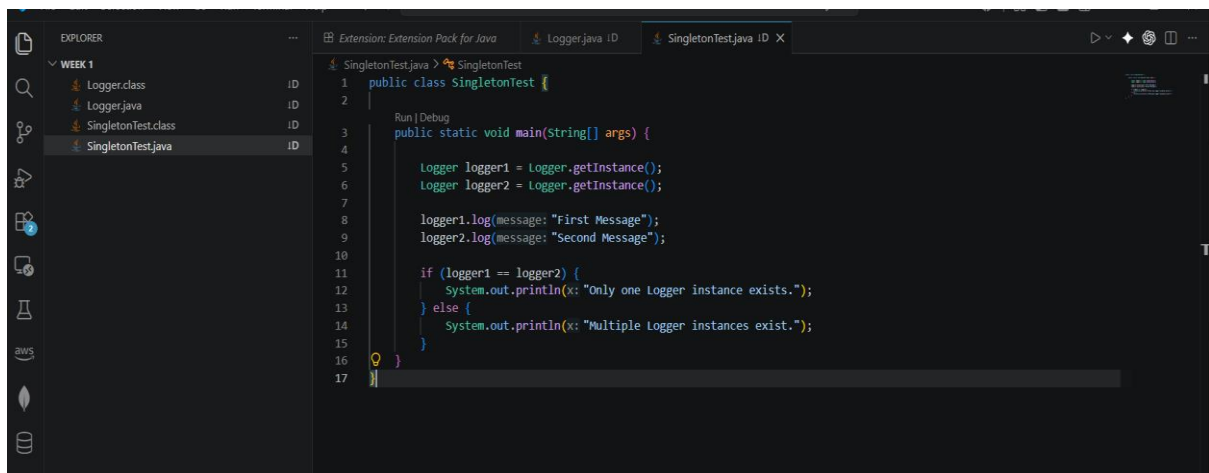
Paste the full code for:

Logger.java



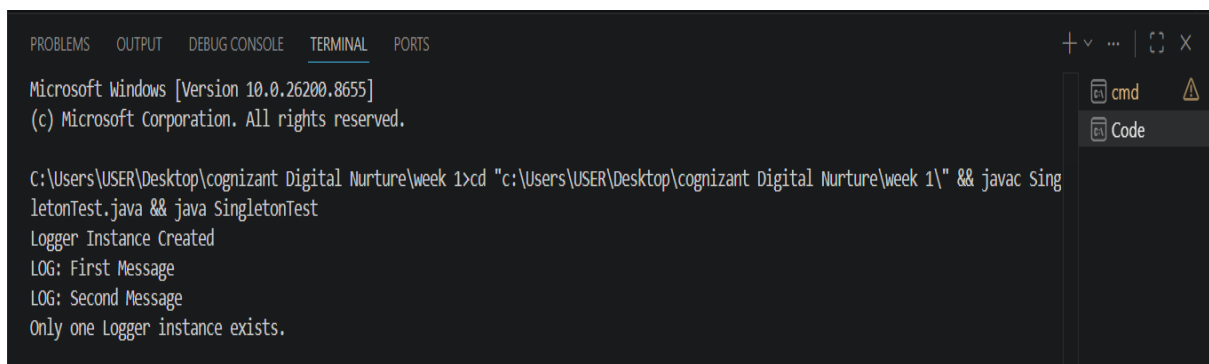
```
1 public class Logger {
2
3     private static Logger instance;
4
5     private Logger() {
6         System.out.println("Logger Instance Created");
7     }
8
9     public static Logger getInstance() {
10
11         if (instance == null) {
12             instance = new Logger();
13         }
14
15         return instance;
16     }
17
18     public void log(String message) {
19         System.out.println("LOG: " + message);
20     }
21 }
```

SingletonTest.java



```
1 public class SingletonTest {
2
3     public static void main(String[] args) {
4
5         Logger logger1 = Logger.getInstance();
6         Logger logger2 = Logger.getInstance();
7
8         logger1.log(message: "First Message");
9         logger2.log(message: "Second Message");
10
11         if (logger1 == logger2) {
12             System.out.println(x: "Only one Logger instance exists.");
13         } else {
14             System.out.println(x: "Multiple Logger instances exist.");
15         }
16     }
17 }
```

OUTPUT:



```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER\Desktop\cognizant Digital Nurture\week 1>cd "c:\Users\USER\Desktop\cognizant Digital Nurture\week 1" && javac SingletonTest.java && java SingletonTest
Logger Instance Created
LOG: First Message
LOG: Second Message
Only one Logger instance exists.
```

Result

The Singleton Pattern was successfully implemented and verified.

Exercise 2: Implementing the Factory Method Pattern

Aim

To implement the Factory Method Design Pattern for creating different types of documents such as Word, PDF, and Excel documents.

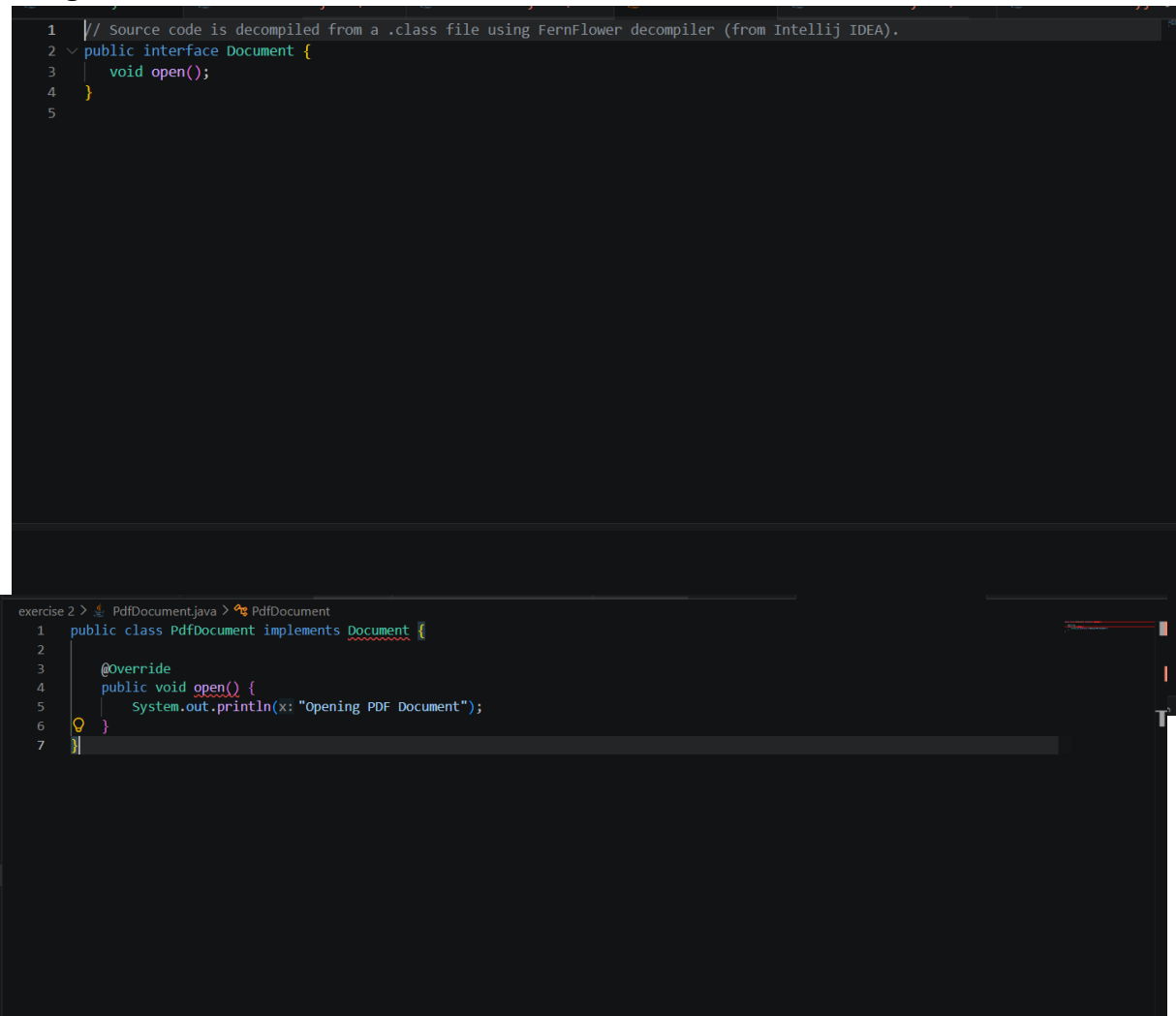
Theory

The Factory Method Pattern is a Creational Design Pattern that provides an interface for creating objects while allowing subclasses or factory methods to determine the type of object to create. This pattern helps in achieving loose coupling between object creation and usage.

Algorithm

1. Create an interface named Document.
2. Define an open() method in the interface.
3. Create WordDocument, PdfDocument, and ExcelDocument classes implementing the Document interface.
4. Create a DocumentFactory class to generate different document objects.
5. Create a test class to demonstrate the creation of various document types.
6. Execute the program and verify the output.

Program



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1 // Source code is decompiled from a .class file using FernFlower decompiler (from IntelliJ IDEA).
2 public interface Document {
3     void open();
4 }
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ExcelDocument.java

```
C:\Users\USER\Desktop\cognizant Digital Nurture\week 1\exercise 2\Document.java • FernFlow decompiler (from IntelliJ IDEA).
Untracked
3 public ExcelDocument() {
4 }
5
6 public void open() {
7     System.out.println("Opening Excel Document");
8 }
9 }
10
```

Factorymethodtest.java

```
1 // Source code is decompiled from a .class file using FernFlow decompiler (from IntelliJ IDEA).
2 public class FactoryMethodTest {
3     public FactoryMethodTest() {
4     }
5
6     public static void main(String[] var0) {
7         DocumentFactory var1 = new DocumentFactory();
8         Document var2 = var1.createDocument("WORD");
9         Document var3 = var1.createDocument("PDF");
10        Document var4 = var1.createDocument("EXCEL");
11        var2.open();
12        var3.open();
13        var4.open();
14    }
15 }
16
```

Output

The screenshot shows the IntelliJ IDEA IDE interface. On the left, the 'EXPLORER' panel displays the project structure for 'WEEK 1', including 'exercise 1' and 'exercise 2'. Under 'exercise 2', several Java files are listed, such as 'Document.class', 'DocumentFactory.class', 'ExcelDocument.class', 'FactoryMethodTest.class', 'PdfDocument.class', 'WordDocument.class', and 'WordDocument.java'. The main editor area shows the source code of 'Document.java', which defines a 'public interface Document' with a 'void open();' method. The 'TERMINAL' panel at the bottom displays the command prompt output, showing the execution of 'javac *.java' and 'java FactoryMethodTest', resulting in the messages 'Opening Word Document', 'Opening PDF Document', and 'Opening Excel Document'.

```
Document.java U X WordDocument.java 2, U PdfDocument.java 2, U ExcelDocument.java 2, U DocumentFactory.java 4, U
exercise 2 > Document.java > Document
1 public interface Document {
2     void open();
3 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
public static void main(String[] args)
or a JavaFX application class must extend javafx.application.Application
c:\Users\USER\Desktop\cognizant Digital Nurture\week 1\exercise 2>cd "c:\Users\USER\Desktop\cognizant Digital Nurture\week 1\exercise 2"
c:\Users\USER\Desktop\cognizant Digital Nurture\week 1\exercise 2>javac *.java
c:\Users\USER\Desktop\cognizant Digital Nurture\week 1\exercise 2>java FactoryMethodTest
Opening Word Document
Opening PDF Document
Opening Excel Document
c:\Users\USER\Desktop\cognizant Digital Nurture\week 1\exercise 2>
```

